Ryan Anselm

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Education

The University of Texas at Austin Sept. 2025 - Present
Ph.D. in Computer Science
Advised by Scott Aaronson and Nick Hunter-Jones
NSF Graduate Research Fellow

Columbia University Sept. 2021 - May 2024
B.S. in Computer Science, *Magna Cum Laude*
Minors in Applied Math & Applied Physics
GPA: 4.05/4.00

Presentations

A Near-Maximal, Verifiable Quantum Advantage in One-Way Communication. Invited Talk. Visa Research, August 2026.

Quantum-inspired Tree Tensor Network Methods for Compression and Transformation of Multivariate Functions. **R. Anselm**, R. Levy, J. Tindall. Tensor Networks and Classical Optimization Session, American Physical Society Global Physics Summit, March 2025.

Machine learning optimal prognostic moments for single-category cloud microphysics parameterizations. **R. Anselm**, M. van Lier-Walqui, K. Lamb. Processes of (Sub) Cloud Scales: Modeling, Observations, and Parameterizations Session, American Geophysical Union Fall Meeting, December 2022.

Research Experience

Flatiron Institute, *Summer Research Assistant* May 2024 - Aug. 2024
Hosted by the Center for Computational Quantum Physics
Supervised by Ryan Levy & Joseph Tindall

- Contributed to a Julia library generalizing the Quantized Tensor Train (QTT) format to arbitrary Tree Tensor Networks (TTNs), a framework for compressed, low-rank representations of multivariate functions.
- Ran large-scale numerical experiments on compute clusters to study function structure vs. optimal tree layout, and implemented efficient TTN operators (e.g., Discrete Fourier Transform, Koopman) for solving differential equations.

Columbia Quantum Computing Theory Group, *Research Assistant* Aug. 2023 - May 2024
Supervised by Henry Yuen

- Studied complexity-theoretic lower bounds for hybrid digital-analog quantum computation and architecture-aware circuit compilation for reconfigurable neutral atom array quantum computers.

Learning the Earth with AI and Physics (LEAP) REU Jun. 2022 - Aug. 2022
Supervised by Kara Lamb, Marcus van Lier-Walqui

- Trained neural network models to emulate cloud microphysics, achieving comparable accuracy to traditional bulk/bin simulation methods. Worked with the ML Python ecosystem (Numpy, Pytorch, Scikit-learn, Scipy, Xarray, etc.) to implement and evaluate models.

UT Austin Computational Materials Freshman Research Initiative Jan. 2021 - Aug. 2021
Supervised by Wenrui Chai

- Developed Python and Fortran algorithms to efficiently locate charge density saddle points for Bader, a computational charge analysis code.

Awards

2025 NSF Graduate Research Fellowship
2024 MIT iQuHack Quantum Hackathon, 1st place in track
2022 Top 350 placement, William Lowell Putnam Mathematical Competition

Work Experience

Salesforce Sept. 2024 - Sept. 2025
Associate Member of Technical Staff

- Built core components of Salesforce's Agentforce agentic reasoner: enabled tool calls into core Salesforce apps and reworked token streaming to support rich UI responses while minimizing latency regressions. Stack: Python, Java, Git/GitHub, CI/CD.

Salesforce May 2023 - Aug. 2023
Software Engineering Intern

Teaching Experience

UT Austin CS358H Intro to Quantum Information Science, *Teaching Assistant*	Fall 2026
Columbia University COMS 4232 Advanced Algorithms, *Teaching Assistant*	Spring 2024
Columbia University COMS 4236 Quantum Computing, *Teaching Assistant*	Fall 2023
Columbia University COMS 3261 Computer Science Theory, *Teaching Assistant*	Spring 2023
Columbia University CSOR 4231 Analysis of Algorithms I, *Teaching Assistant*	Fall 2022

Outreach & Service

Columbia University Science Olympiad, *Co-President & Co-Founder* 2021 - 2024

- Co-founded an undergraduate student group and led a team of 12 undergraduates to organize an annual Science Olympiad invitational tournament hosted at Columbia University for 500+ high school students.
- Authored 12+ Science Olympiad tests as an volunteer for several tournaments including the 2021 national tournament, Columbia, MIT, UT Austin, UCincinnati, and UPenn invitationals.

Columbia University Undergraduate Learning Seminar in Theoretical Computer Science 2022-2024

- Organized a reading group on quantum complexity theory and presented on foundational papers.

Selected Coursework

- **Computer Science:** Quantum Information Science I and II, Quantum Protocols, Computational Complexity, Advanced Algorithms, Approximation Algorithms, Computational Learning Theory, Unconditional Lower Bounds & Derandomization, Machine Learning, Unsupervised Learning, Operating Systems, Compilers
- **Mathematics:** Real Analysis, Applied Functional Analysis, Partial Differential Equations
- **Physics:** Quantum Mechanics I and II, Quantum Optics, Thermodynamics & Statistical Mechanics, Applied Electrodynamics, Classical Mechanics

Skills

- **Programming Languages:** Python, Java, Julia, C, C++
- **Technologies:** Git, PyTorch, NumPy, Qiskit, Mathematica, LaTeX
- **Technical Areas:** Quantum Computing, Theoretical Computer Science, Tensor Networks, Machine Learning